

Climate Exploration Platform

Interactive visualization of long-term climate trends and extreme events

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Motivation

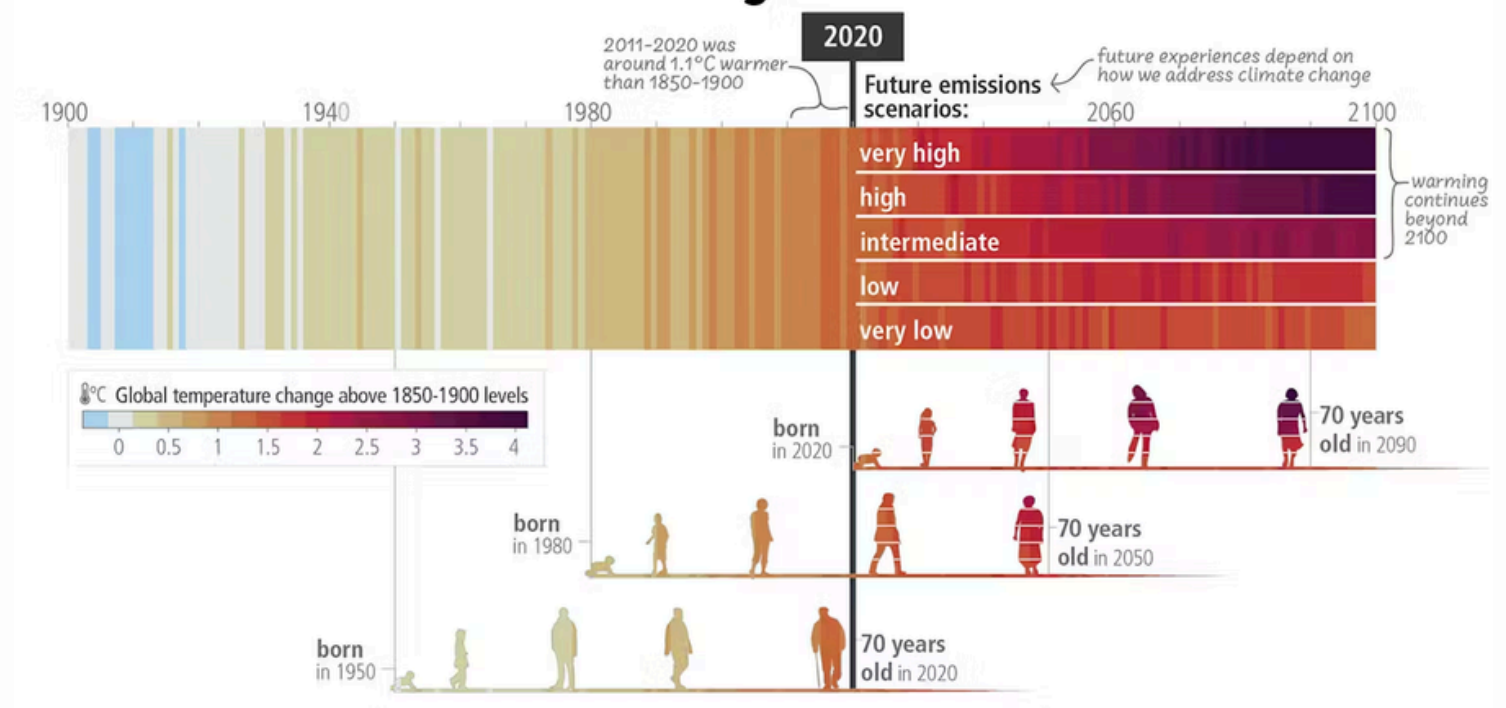
Climate change impacts regions and countries unevenly.

Extreme weather events such as droughts and wildfires have long-term environmental and socio-economic consequences.

Large amounts of climate data exist, but:

- they are fragmented
- difficult to compare across regions
- hard to interpret without interactive visualization

Just how bad climate change is for current and future generations depends on choices we make now – or very soon



<https://citynews.com.au/2023/climate-report-the-world-is-in-deep-trouble/>

Project Goal

Visualization

Track long-term climate trends and highlight extreme weather events globally

Large Display

Optimized for high-resolution screens with synchronized side-by-side dashboards.

Exploration

No scrolling required. Full overview with detailed interactive visualizations.

Data

Multiple data sources were merged for each dashboard:

- Renewable Share: renewable energy, CO₂, GDP, population.
- Drought Index: crop yields, food expenditure per capita, drought events, temperature.
- Wildfire: temperature and fire occurrence data.

Data cleaning included:

- Removing duplicates
- Handling missing values
- Standardizing country codes
- Clustering

Problems / Challenges

Heterogeneous data sources:

- climate data
- extreme weather event data
- socio-economic indicators

Different temporal resolutions (yearly vs. monthly)

Temporal misalignment between climate and agriculture data (no overlapping periods)

Incomplete and sparse datasets

Large data volume leading to computational constraints (CPU limitations)

Design Decisions

Key Design Decisions

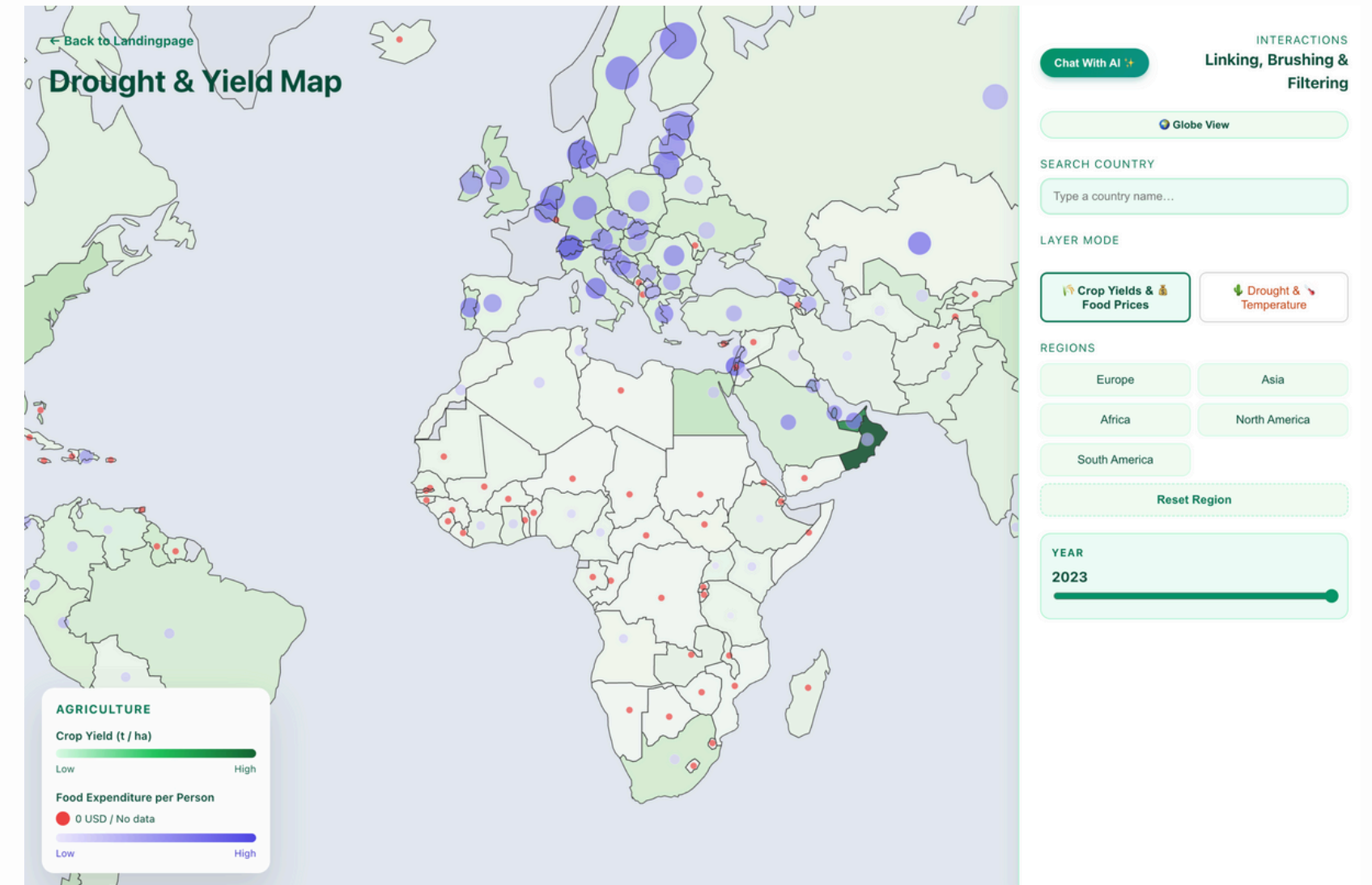
- Topic-specific dashboards:
 - Drought index
 - Wildfire
 - Renewable Share
- Reduces cognitive load and improves clarity
- Interactive visualizations instead of static charts
- Side-by-side analysis

Design Approach

- Focus on exploratory analysis rather than static reporting
- Discover patterns
- Compare regions
- Analyze trends over time

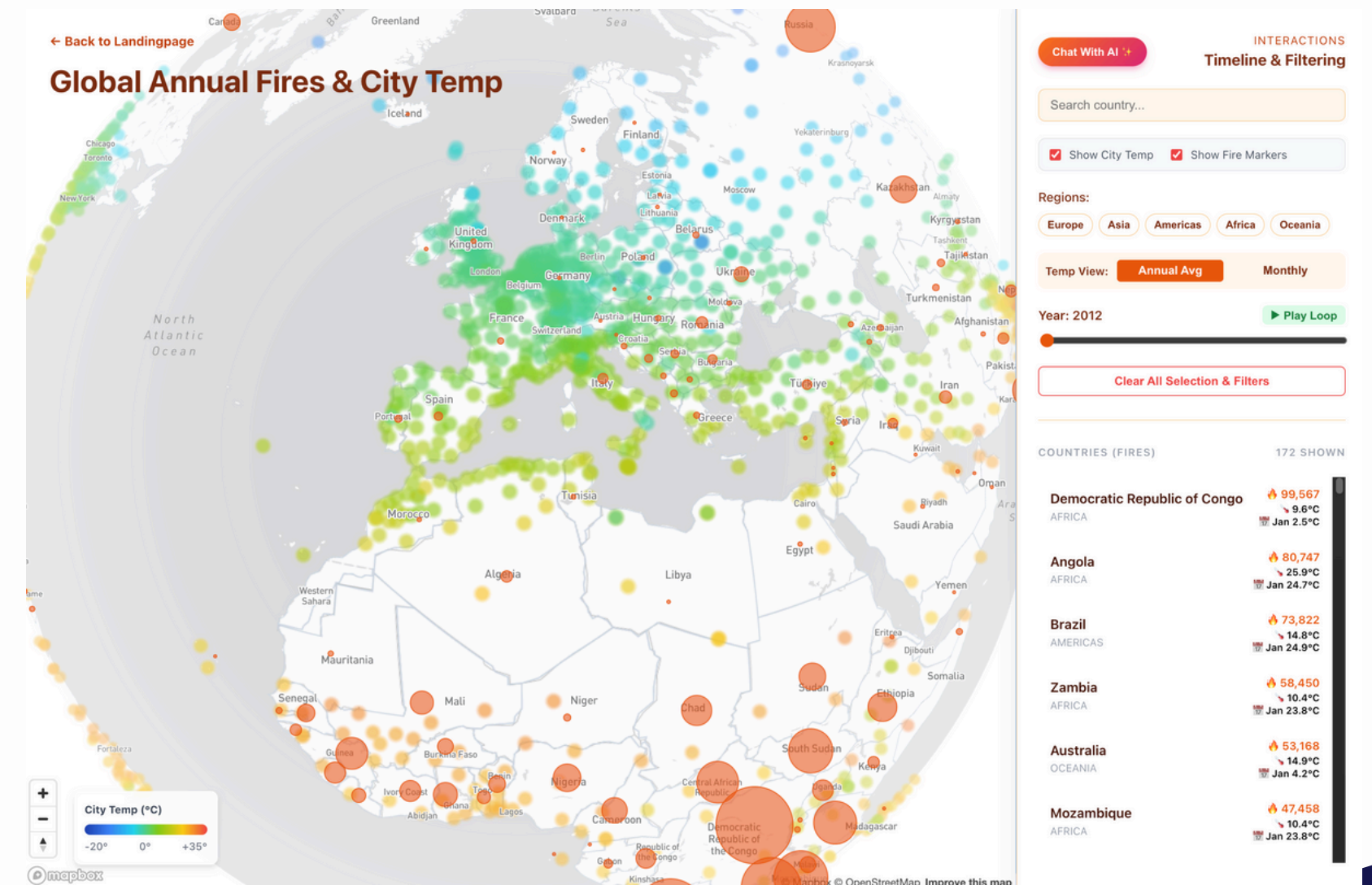
Drought Index Dashboard

- Layer Mode: switch between crop yields & food expenditure or drought events & temperature
- Multiple Views: switch between Map and Globe
- Region Selection: filter by continent or country
- Detailed Insights: hover/click to view drought details (intensity, severity, duration, score) or yield/expenditure details
- Interactive AI Assistant: provides context-aware explanations for the current dataset



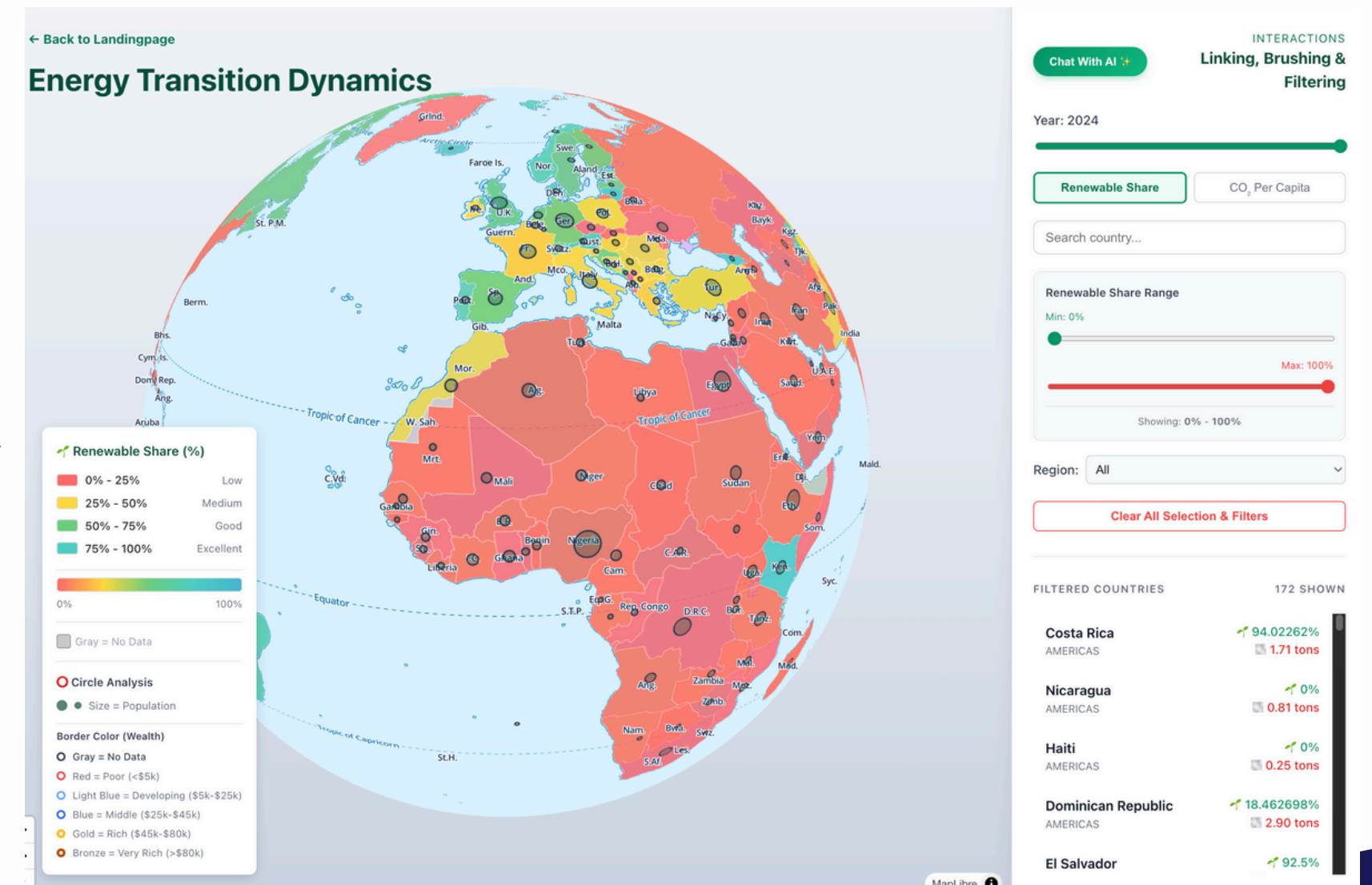
Wildfire Index Dashboard

- Layer Mode: explore temperature patterns alongside wildfire occurrences
- Multiple Views: switch between Year and Month based.
- Region Selection: filter by continent or country
- Detailed Insights: hover/click to view fire frequency and temperature information
- Interactive AI Assistant: provides context-aware explanations for the current dataset



Renewable Energy Dashboard

- Layer Mode: Switch between Renewable Energy Share (%) and CO₂ Emissions per Capita (tons).
- Visual Encoding: Color fill , Circle size , Circle Border Color
- Region Selection: filter by continent or country
- Range Filters: Min/max sliders
- Detailed Insights: Hover click to view Renewable Share, CO₂ , Total GDP, and Population
- Interactive AI Assistant: Provides context-sensitive explanations for energy transition trends and anomalies.



Side-By-Side Dashboard

Comparative Analysis

Enables users to view two different datasets or themes simultaneously

Flexible Theme Selection

Users can pick any combination of topics (e.g., drought, wildfire, renewable energy) for comparison.

Synchronized View

When the same theme is selected on both sides, view and timeline can be synced for direct comparison.

Limitations

- Temporal Gaps: climate and agriculture datasets cover different time periods, limiting direct trend comparisons.
- Interaction Limits: side-by-side comparisons are only available for specific data combinations.
- Computational Constraints: large data volume led to CPU limitations, restricting model complexity and real-time processing.
- Data Accessibility: challenges in accessing, integrating, and harmonizing data from multiple external sources.
- Frontend Dependency: the frontend is optimized specifically for the Powerwall unit; accessing or modifying it from other devices (e.g., desktop, tablet, or mobile) may lead to scaling issues and inconsistent behavior.



THANK YOU!

QUESTIONS?

